

- 2 *Anabaena doliolum* is a species of photosynthetic prokaryote that is found in many freshwater ecosystems. The individual cells of *A. doliolum* form long chains.

Figure 2.1 shows a photomicrograph of several chains of *A. doliolum*.

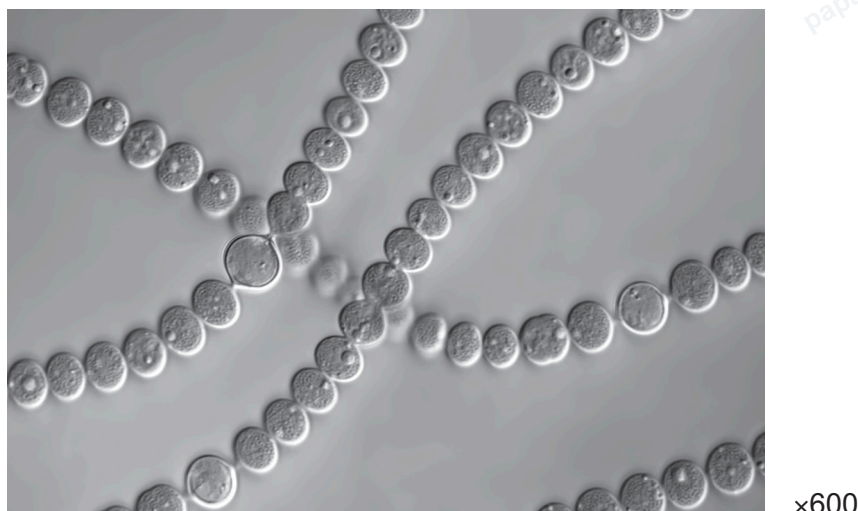


Figure 2.1

Some scientists investigated the effect of sodium chloride concentration on the rate of photosynthesis of *A. doliolum* by carrying out two different experiments.

In the first experiment, the scientists estimated the rate of photosynthesis of *A. doliolum* by determining the rate of oxygen production.

- A sample of *A. doliolum* cells was placed in a set volume of 0.25 mol dm^{-3} solution of sodium chloride.
- An oxygen electrode was used to determine the rate of oxygen production.
- The procedure was repeated with 0.20, 0.15, 0.10, 0.05 and 0.00 mol dm^{-3} solutions of sodium chloride.

In the second experiment, the scientists estimated the rate of photosynthesis of *A. doliolum* by determining the rate of DCPIP reduction. DCPIP changes colour from blue to colourless when it is reduced.

- An enzyme was added to a sample of *A. doliolum* cells. The enzyme partially hydrolysed the bacterial cell walls of *A. doliolum*.
- The sample of *A. doliolum* cells was placed in a test-tube with a set volume of 0.25 mol dm^{-3} solution of sodium chloride. This mixture was green in colour.
- A set volume of DCPIP was added to the test-tube to give a blue-green colour.
- The time taken for the mixture in the test-tube to change colour from blue-green to green was measured and the rate of DCPIP reduction was determined. The rate of reduction of DCPIP is a measure of the rate of NADP reduction during the light-dependent stage of photosynthesis.
- The procedure was repeated with 0.20, 0.15, 0.10, 0.05 and 0.00 mol dm^{-3} solutions of sodium chloride.



(a) Identify the **two dependent** variables in this investigation.

first experiment

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second experiment

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[2]

(b) The scientists standardised the light intensity, temperature and volumes of solutions in the first and second experiments.

Identify **one other** variable that should be standardised in the **second** experiment.

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..... [1]

(c) Suggest a reason why an enzyme was added to partially hydrolyse the bacterial cell walls of *A. doliolum* cells before measuring the rate of DCPIP reduction.

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..... [1]

- (d) The scientists were given a 0.50 mol dm^{-3} stock solution of sodium chloride. The scientists used the stock solution to make sodium chloride solutions with concentrations of 0.25, 0.20, 0.15, 0.10, 0.05 and 0.00 mol dm^{-3} .

The scientists made 20 cm^3 of each solution in separate beakers.

Complete Table 2.1 to show how 20 cm^3 of these solutions could be made by **proportional** dilution of the 0.50 mol dm^{-3} stock solution of sodium chloride.

Table 2.1

concentration of sodium chloride solution / mol dm^{-3}	volume of 0.50 mol dm^{-3} stock solution of sodium chloride / cm^3	volume of / cm^3
0.25		
0.20		
0.15		
0.10		
0.05		
0.00		

[2]



Turn over

(e) Figure 2.2 shows the results of the investigation.

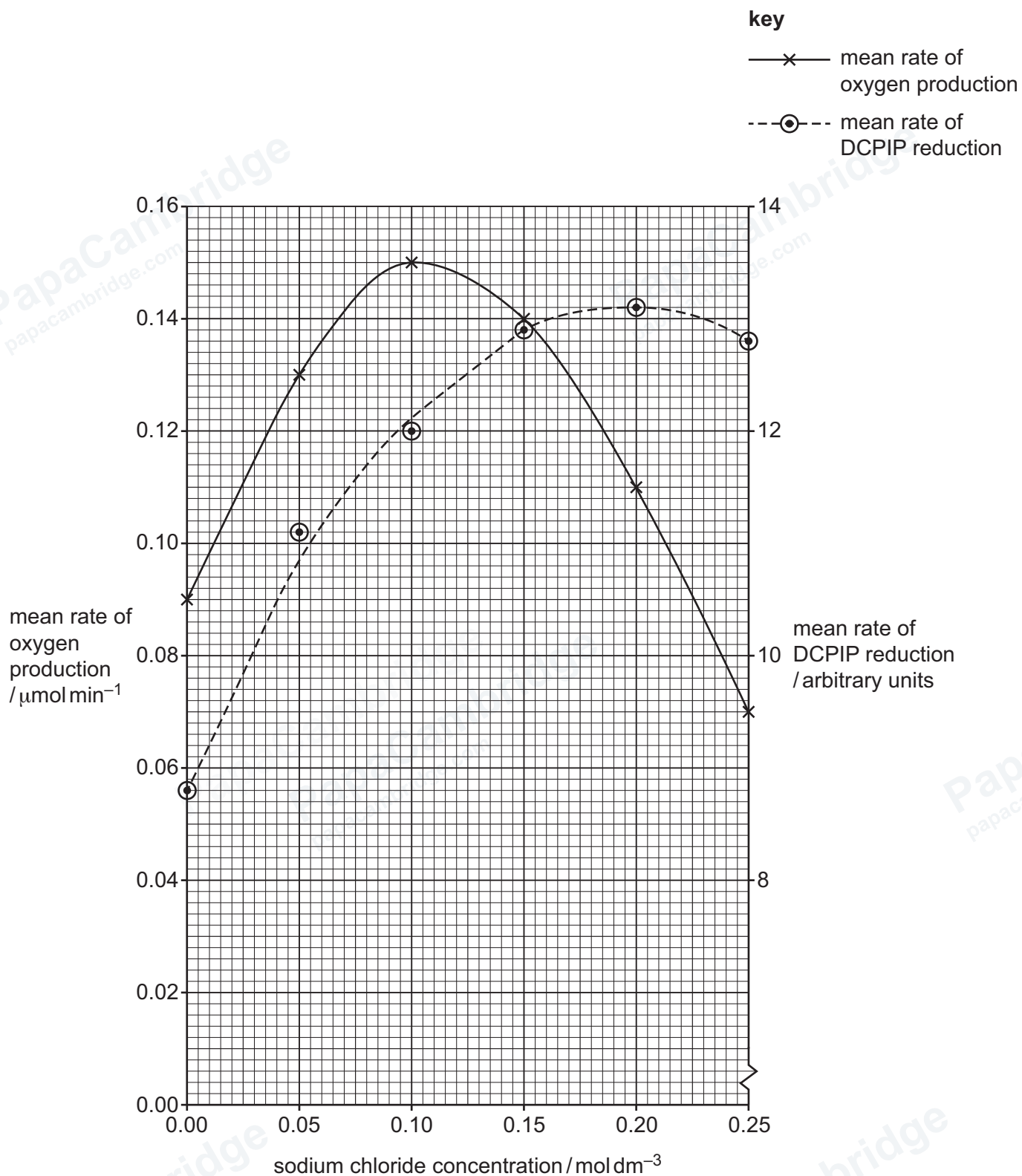


Figure 2.2



Use Figure 2.2 to identify the key differences between the two sets of data from the investigation.

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..... [2]

[Total: 8]